Example 6.3

$$X_t = \theta Z_{t-1} + Z_t$$

$$\rho(0) = 1$$

$$\rho(1) = \rho(-1) = \frac{\theta}{1+\theta^2} \quad \text{(Lecture 3)}$$

$$\rho(h) = 0 \quad \text{for } |h| \geq 2.$$

For the PACF

$$\rho(0,1) = \rho(1) = \frac{\theta}{1+\theta^2}$$

$$\rho(0,2|1) = \frac{\rho_2 - \rho_1^2}{1-\rho_1^2}$$

$$= \frac{-\dfrac{\theta^2}{(1+\theta^2)^2}}{1 - \dfrac{\theta^2}{(1+\theta^2)^2}}$$

$$= \frac{-\dfrac{\theta^2}{(1+\theta^2)^2}}{\dfrac{(1+\theta^2)^2 - \theta^2}{(1+\theta^2)^2}}$$

$$= \frac{-\theta^2}{(1+\theta^2)^2 - \theta^2}$$

$$= \frac{-\theta^2}{1+\theta^2+\theta^4}$$

For $\rho(0,3 \mid 1,2)$ we seek $\beta_{3,3}$ where

$$P_3 \begin{pmatrix} \beta_{3,1} \\ \beta_{3,2} \\ \beta_{3,3} \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{pmatrix}$$

$$P_3 = \begin{pmatrix} 1 & \rho_1 & \rho_2 \\ \rho_1 & 1 & \rho_1 \\ \rho_2 & \rho_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \rho_1 & 0 \\ \rho_1 & 1 & \rho_1 \\ 0 & \rho_1 & 1 \end{pmatrix}$$

Use Cramer's rule $\quad \beta_{3,3} = \dfrac{\det P_3^*}{\det P_3}$

$$P_3^* = \begin{pmatrix} 1 & \rho_1 & \rho_1 \\ \rho_1 & 1 & \rho_2 \\ 0 & \rho_1 & \rho_3 \end{pmatrix} = \begin{pmatrix} 1 & \rho_1 & \rho_1 \\ \rho_1 & 1 & 0 \\ 0 & \rho_1 & 0 \end{pmatrix}$$

$$\det P_3 = 1 - \rho_1^2 - \rho_1^2 = 1 - 2\rho_1^2$$

$$\det P_3^* = \rho_1^3$$

$$\det P_3 = 1 - \frac{2\theta^2}{(1+\theta^2)^2} = \frac{(1+\theta^2)^2 - 2\theta^2}{(1+\theta^2)^2}$$

$$= \frac{1+\theta^4}{(1+\theta^2)^2}$$

$$\det P_3^* = \frac{\theta^3}{(1+\theta^2)^3}$$

$$\beta_{3,3} = \frac{\det P_3^*}{\det P_3}$$

$$= \frac{\dfrac{\theta^3}{(1+\theta^2)^3}}{\dfrac{1+\theta^4}{(1+\theta^2)^2}}$$

$$= \frac{\left[\dfrac{\theta^3}{\cancel{(1+\theta^2)^3}}\right]}{\dfrac{(1+\theta^4)(1+\theta^2)}{\cancel{(1+\theta^2)^3}}}$$

$$= \frac{\theta^3}{(1+\theta^4)(1+\theta^2)}$$

$$= \frac{\theta^3}{1+\theta^2+\theta^4+\theta^6}$$

Generally

$$\rho(0,\ldots,m \mid 1,\ldots,m-1) = \frac{(-1)^{m+1}\,\theta^m}{\sum_{j=0}^{m} \theta^{2j}}$$